



SYMBIOSIS INSTITUTE OF TECHNOLOGY

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NAAC with “A” Grade

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Batch – B3

COURSE – JAVA

Experiment 11-

Write a Java program to demonstrate the implementation of interfaces

AIM – Write a Java program to demonstrate the implementation of interfaces

Program 11

```
interface Vehicle {  
    void start();
```

```

    void stop();
}

// Car class implementing Vehicle interface
class Car implements Vehicle {
    public void start() {
        System.out.println("Car is starting.");
    }
    public void stop() {
        System.out.println("Car has stopped.");
    }
}

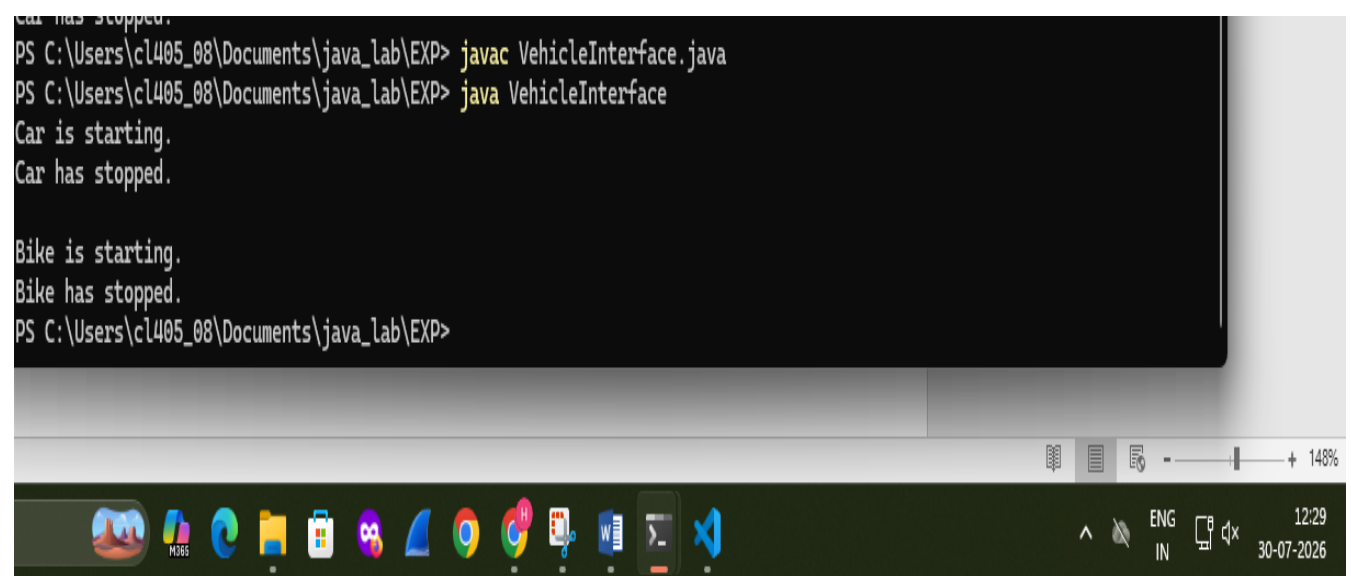
// Bike class implementing Vehicle interface
class Bike implements Vehicle {
    public void start() {
        System.out.println("Bike is starting.");
    }
    public void stop() {
        System.out.println("Bike has stopped.");
    }
}

// Main class
public class VehicleInterface {
    public static void main(String[] args) {
        // Interface reference for Car
        Vehicle car = new Car();
        car.start();
        car.stop();
        System.out.println();
    }
}

```

```
// Interface reference for Bike  
  
Vehicle bike = new Bike();  
  
bike.start();  
  
bike.stop();  
}  
}
```

Output



```
car has stopped.  
PS C:\Users\cl405_08\Documents\java_lab\EXP> javac VehicleInterface.java  
PS C:\Users\cl405_08\Documents\java_lab\EXP> java VehicleInterface  
Car is starting.  
Car has stopped.  
  
Bike is starting.  
Bike has stopped.  
PS C:\Users\cl405_08\Documents\java_lab\EXP>
```

EXERCISE

11.1 . Create an interface Printable and implement it in Student and Employee classes.

Program 11.1 -

```
// Interface
interface Printable {

    // Abstract method
    void printDetails();
}

// Student class
class Student implements Printable {

    // Implementing interface method
    public void printDetails() {
        System.out.println("Student Name : Sourabh Gupta");
        System.out.println("Roll Number : 101");
    }
}

// Employee class
class Employee implements Printable {

    // Implementing interface method
    public void printDetails() {
        System.out.println("Employee Name : Sourabh Gupta");
        System.out.println("Employee ID : 501");
    }
}

// Main class
public class Main {

    public static void main(String[] args) {
```

```

// Creating objects
Student s = new Student();
Employee e = new Employee();

// Calling methods
System.out.println("Student Details:");
s.printDetails();

System.out.println();

System.out.println("Employee Details:");
e.printDetails();
}
}

```

Screenshot 11.1 –

```

Student Details:
Student Name : Sourabh Gupta
Roll Number  : 101

Employee Details:
Employee Name : Sourabh Gupta
Employee ID   : 501
PS C:\Users\cl404_17\javav\java> |

```

Program 11.2- Create an interface Switchable with a method turnOn(). Implement the interface in Light and Fan classes to display the device status.

```
interface Switchable {
    void turnOn();
}

// Light class implementing Switchable interface
class Light implements Switchable {

    public void turnOn() {
        System.out.println("Light is turned ON.");
    }
}

// Fan class implementing Switchable interface
class Fan implements Switchable {

    public void turnOn() {
        System.out.println("Fan is turned ON.");
    }
}

// Main class
public class SwitchableDemo {

    public static void main(String[] args) {

        // Interface reference for Light
        Switchable light = new Light();
        light.turnOn();

        System.out.println();

        // Interface reference for Fan
        Switchable fan = new Fan();
        fan.turnOn();
    }
}
```

Screenshot 11.2 –

```
PS C:\Users\cl404_17\javav\java> javac SwitchableDemo.  
PS C:\Users\cl404_17\javav\java> java SwitchableDemo  
Light is turned ON.  
  
Fan is turned ON.  
PS C:\Users\cl404_17\javav\java> |
```